Annex B — Data Preparation, Analysis & Visualisation

Tan I-Je 252426C Individual Submission (Person 2: Quality of Life & Social Indicators)

1. Focus Area and Analytical Objective

Focus Area

Quality of Life & Social Indicators

Guiding Question

> How liveable is Singapore beyond income?

While Singapore consistently performs well on economic indicators, income alone does not fully capture residents' lived experiences. This analysis evaluates Singapore's quality of life using a multi-dimensional framework encompassing **health, education, safety, and subjective well-being**, benchmarked against regional and global peers. The objective is to assess whether strong institutional and economic performance translates into broader social outcomes that contribute to long-term liveability and sustainability.

2. Data Sources and Indicators

The analysis integrates multiple authoritative global datasets to ensure comparability, reliability, and relevance to the research question.

Focus Area	Dataset / Indicator	Source	URL
Health	Life Expectancy at Birth	World Bank	https://data.worldbank.org/indicator/SP.DYN.LE00.IN
Health	Current Health Expenditure per Capita (PPP)	World Bank	https://data.worldbank.org/indicator/SH.XPD.CHEX.PP.CD
Education	Educational Attainment (Population 25+, at least upper secondary, %)	World Bank	https://data.worldbank.org/indicator/SE.SEC.CUAT.UP.ZS
Safety	Homicide Rate (per 100,000	Our World in Data (UNODC)	https://ourworldindata.org/search?q=Homicide+rate&countries=Singapore&resultType=all

Focus Area	Dataset / Indicator	Source	URL
	population)		
Well-being	Life Satisfaction	World Bank	https://ourworldindata.org/grapher/happiness-cantril-ladder
Well-being	Composite Quality of Life Index	Derived	Constructed in Power BI

These datasets were selected as they are widely used in international benchmarking studies and provide consistent definitions across countries.

3. Data Preparation and Transformation

3.1 Data Cleansing

Raw datasets were imported into Power BI using Power Query and subjected to systematic cleansing procedures:

- Rows containing **missing values** in *health expenditure per capita (PPP)* and *life expectancy* were removed, as these represented years with unavailable data and would distort cross-country and time-series comparisons.
- Duplicate records and non-country entities unrelated to the analysis were excluded.
- Column names and data types were standardised to ensure consistency across datasets.

These steps ensured that all retained observations were complete, comparable, and analytically valid.

3.2 Indicator Selection and Refinement

During exploratory analysis, **upper secondary completion rate (%)** was initially reviewed as a potential education indicator. However, it was excluded from the final analysis due to **incomplete and inconsistent coverage** across the selected countries and years.

Instead, **educational attainment of the population aged 25 and above (at least upper secondary)** was retained, as it:

- Provides a more stable long-term measure of education outcomes,
- Reflects cumulative educational achievement rather than short-term cohort effects,
- Enables fairer cross-country comparison.

This decision aligns with best practices in social indicator analysis.

3.3 Benchmark Groups and Aggregates

In addition to individual countries, **regional and income-group aggregates** from Our World in Data were retained:

- Singapore
- Asia
- High-income countries
- World

These aggregates serve as meaningful benchmarks, allowing the analysis to answer not only *how Singapore performs*, but *relative to whom*. This supports a broader sustainability and liveability narrative by situating Singapore within both regional and global contexts.

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4. Data Transformation and Normalisation

4.1 Base Metrics

For each indicator, average values were computed to provide descriptive summaries for comparison:

- Average life expectancy
- Average healthcare spending per capita
- Average education attainment
- Average homicide rate
- Average life satisfaction

These metrics form the foundation for subsequent transformation and visualisation.

4.2 Normalisation of Indicators

As the indicators are measured in different units (years, percentages, monetary values, rates), **min–max normalisation** was applied to convert all metrics to a **0–100 scale**:

$$\text{Normalised Score} = \frac{x - \min(x)}{\max(x) - \min(x)} \times 100$$

This approach preserves relative differences while enabling meaningful comparison across dimensions.

Safety Indicator Adjustment

Homicide rate is a *negative indicator* where lower values indicate better outcomes. After normalisation, the score was inverted so that higher values consistently represent better performance:

$$\text{Safety Score} = (1 - \text{Normalised Homicide Rate}) \times 100$$

This ensures interpretive consistency across all quality-of-life dimensions.

5. Construction of the Composite Quality of Life Index

To provide an overall benchmark of liveability, a **composite Quality of Life (QoL) Index** was constructed by averaging the four normalised dimensions:

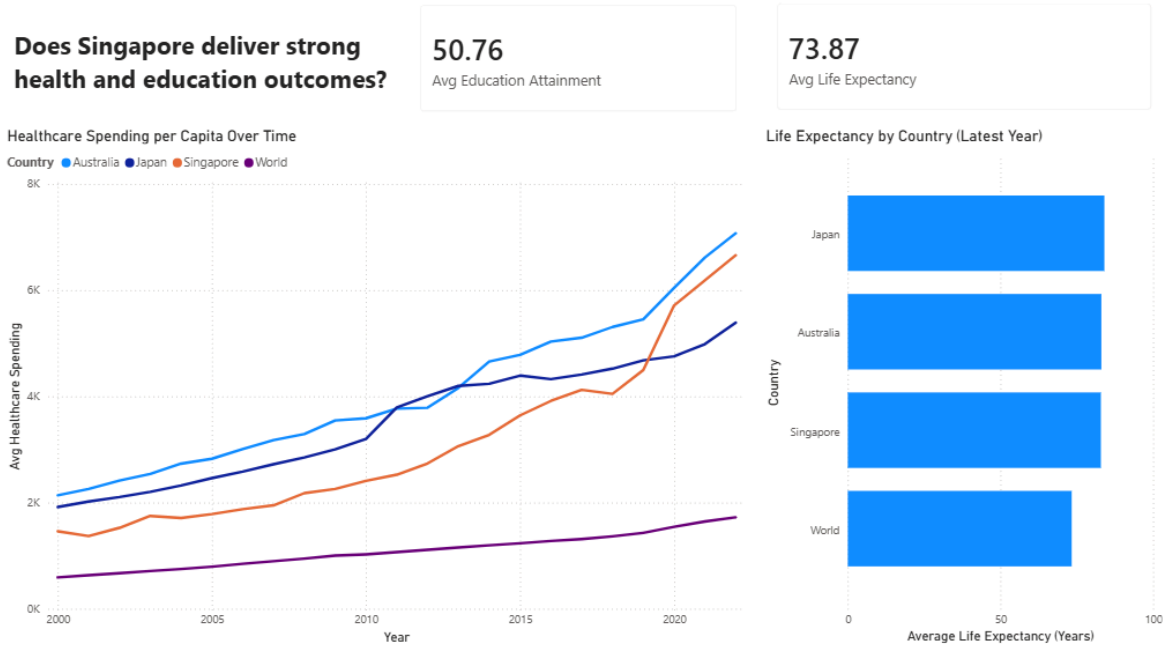
- Health
- Education
- Safety
- Life satisfaction

Each dimension was assigned **equal weight**, reflecting the assumption that all aspects contribute equally to overall quality of life.

This composite index allows for a concise comparison of overall performance while maintaining transparency in its construction.

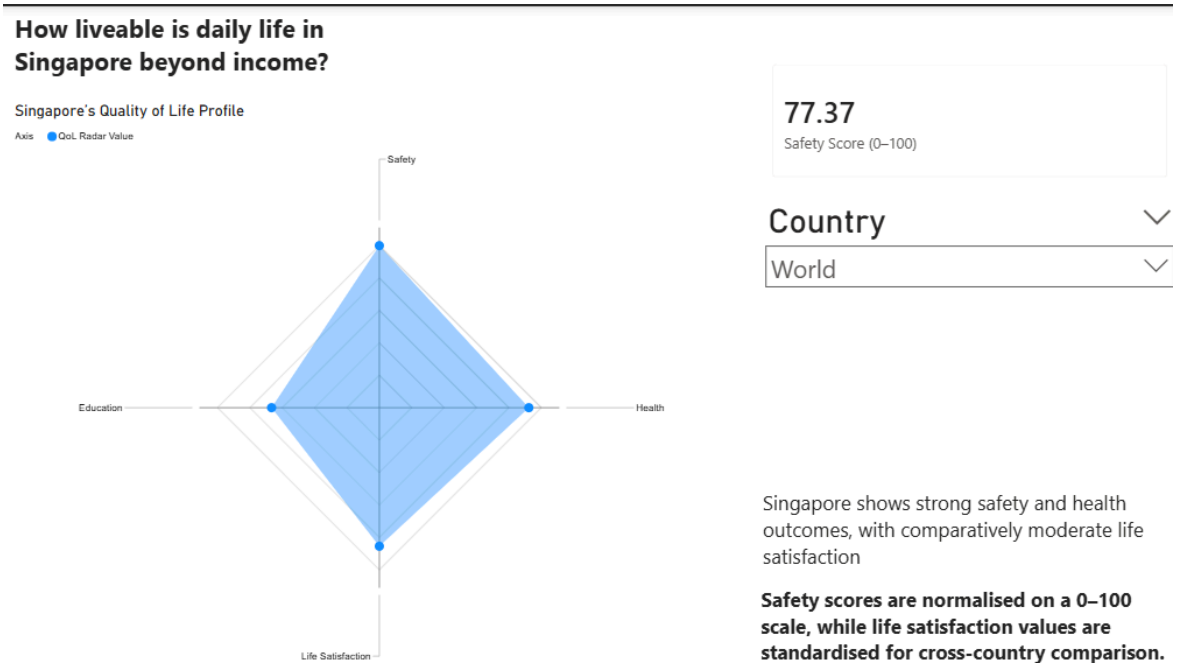
6. Data Analysis and Visualisation

6.1 Health and Education Outcomes



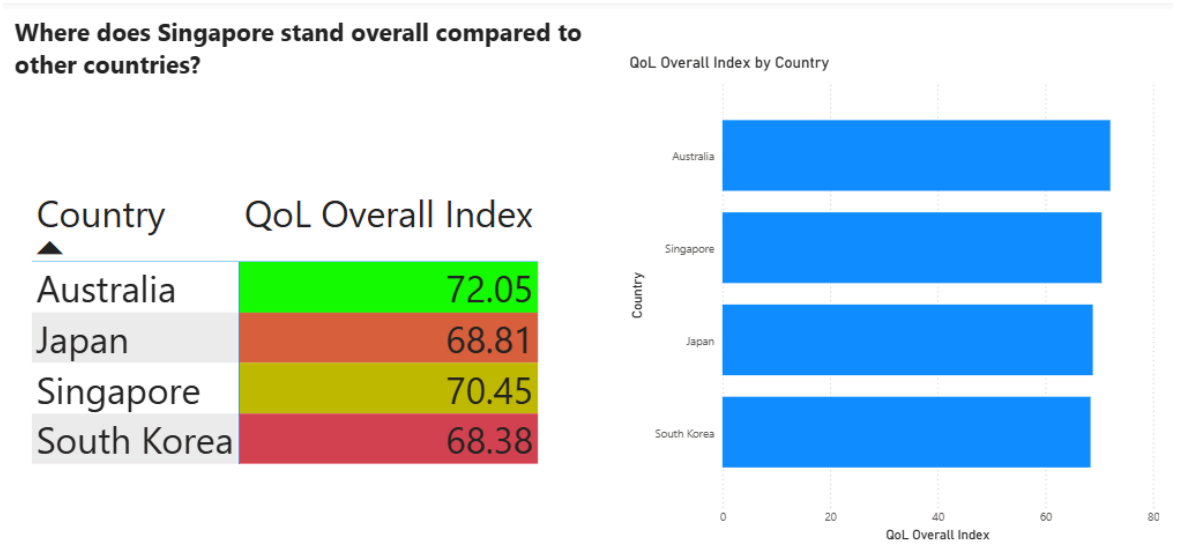
- Healthcare spending trends and life expectancy are benchmarked against peers.
- Education attainment summarises long-run outcomes.
- Singapore performs strongly on health outcomes relative to regional and global benchmarks.

6.2 Social Well-being and Safety



- Radar chart compares the four normalised dimensions.
- Singapore is strongest in **health and safety**.

6.3 Overall Quality of Life Benchmarking



- Heatmap and ranking table compare composite QoL across countries.

- Overall score is moderated by relatively lower life satisfaction.

7. Key Insights

- Singapore demonstrates **strong performance in health and safety**, reflecting effective public systems and governance.
- Education outcomes remain robust, supporting long-term human capital development.
- Subjective well-being moderates Singapore's overall quality-of-life ranking, suggesting that high income and strong infrastructure do not fully translate into higher life satisfaction.
- Benchmarking against regional and income-group aggregates provides valuable context for evaluating sustainability and long-term liveability.

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8. Conclusion

Singapore remains highly liveable beyond income, with particularly strong health, safety, and education outcomes. Subjective well-being is a key moderating factor in overall QoL, highlighting important trade-offs for long-term sustainability planning.